

Exact to the Quarter, Lost in the Generation: US AI-Chip Export Controls in Product-Level Shipment Panels

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analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

At product granularity, the US export-control rounds on AI accelerators localize in time but not in attribution. In chip $\times$ quarter panels of shipment flows built from Epoch AI’s chip-sales and chip-components estimates, all six known-treatment difference-in-differences scans put the maximum-likelihood-ratio break at the declared policy quarter *exactly*, and the October-2023 A800/H800 ban clears significance (Bernoulli scan $p = 0.030$; the banned SKUs’ shipments go to literal zero within one quarter), while the April-2025 H20 licensing round localizes exactly but insignificantly ($p = 0.44$ – 0.60 on sales, 0.13 – 0.23 on components). The H20 effect magnitude is large and sign-correct — $\hat{\tau} = -2.756$ asinh units under two-way fixed effects, CR1 $t = -4.23$, 0 of 6 matched placebos exceeding, at the exact placebo-in-space floor $p = 0.143$ — but cannot be causally attributed: an unsupervised rediscovery scan shows that the panel’s sharpest structure at 2025Q2 is the *entire Hopper-generation wind-down* with the H20 inside it, and the placebo battery generates same-size effects from ordinary product-lifecycle churn, with the matched donor pool either contaminated (3 of 6 controls are Chinese substitutes whose demand the ban plausibly boosted) or too small (3 clean controls, placebo $p = 0.75$). The date localization of the October-2023 round is the headline positive result; the correctly identified lifecycle confound and the pipeline’s clean refusals are the rest of the story.

1 Introduction

On 7 October 2022 the US Bureau of Industry and Security (BIS) imposed the first broad export controls on advanced AI accelerators destined for China, a policy contemporaries called a landmark in US–China technology competition [1]. Nvidia responded within weeks with China-market SKUs — the A800 and H800 — engineered just under the control thresholds. On 17 October 2023 BIS closed that loophole: the updated advanced-computing rule [2] adjusted the chip-level performance parameters so that the workaround SKUs themselves became controlled. Nvidia iterated once more with the H20, a Hopper-generation part compliant with the 2023 thresholds — until 9 April 2025, when the US government informed the company that H20 exports to China (and any chip matching its bandwidth profile) would require a license, indefinitely; Nvidia’s securities filing disclosed up to \$5.5 billion of associated charges [13].

A growing empirical literature evaluates these controls. On the supplier side, [6] show that US firms subject to export controls halt sales to their Chinese customers as intended but largely fail to

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replace them, losing market value, profitability, and employment. On the evasion side, [9] document underground markets for controlled accelerators and estimate the 2023 smuggling flow at hundreds to low thousands of chips. A companion country-level study in this collection [11] found that the October-2023 round *bent* China’s legal chip-stock growth downward while aggregate compute was substantially bypassed. What that literature leaves untested is the sharpest granularity at which the policy is written: the named SKU. Each control round names specific products; if the rounds bind, product-level shipment panels should carry a break at the declared quarter in exactly the named SKUs — and nowhere else.

This paper runs that test three ways with the `natex` toolkit [12]. Leg 1 asks whether a scan-based difference-in-differences that searches over break dates [10], *knowing* which chips were treated, recovers the declared policy quarter from the data. Leg 2 removes the treatment labels entirely and asks what the sharpest structure in each panel actually is. Leg 3 estimates effect magnitudes with the placebo-based inference appropriate to a handful of treated clusters [5, 3]. The answer pattern — exact localization everywhere, significance only for October 2023, attribution defeated by product-lifecycle churn — is documented below, refusals included.

2 Data

Two Epoch AI datasets are used. *Data on AI Chip Sales* [8] provides quarterly shipment-flow estimates by chip model in thousands of H100-equivalents (kH100e); the extract’s estimates were generated in January 2026 with 2025Q4 the last full-coverage quarter. *Data on AI Chip Components* [7] provides quarterly CoWoS advanced-packaging wafer estimates by chip for 2024Q1–2025Q4. From these, three chip $\times$ quarter panels are built (time index $t = 4\text{year} + \text{quarter} - 1$, so $8095 = 2023\text{Q4}$ and $8101 = 2025\text{Q2}$): (i) *sales Oct-23*, 7 chips $\times$ 16 quarters (2022Q1–2025Q4), treated $\{\text{A800, H800}\}$, declared $t_0 = 2023\text{Q4}$; (ii) *sales H20*, 20 chips $\times$ 8 quarters (2024Q1–2025Q4), treated $\{\text{H20}\}$, declared $t_0 = 2025\text{Q2}$; and (iii) *components H20*, 15 chips $\times$ 8 quarters of CoWoS wafers, treated $\{\text{H20, MI308X}\}$, declared $t_0 = 2025\text{Q2}$.

The outcome is asinh of the flow. Zeros are true zeros (no shipments), not missingness; chips born after $t_0 - 1$ are excluded at build time so that treatment cannot be mechanically confounded with birth; and the ragged 2026 tail is dropped at the 2025Q4 snapshot. Because the treatment effect here operates on the extensive margin — banned SKUs’ shipments go to approximately zero — asinh magnitudes are unit-dependent and should not be read as percentage effects [4]; the qualitative reading (shipments driven to zero) is the transform-robust statement. One measurement caveat is carried through the paper: Huawei Ascend and Cambricon Siyuan flows in the source data are annual allocations smeared uniformly across quarters, which makes those series smooth by construction.

3 Design and Methods

Leg 1: known-treatment scans (date localization). For each panel, the `natex` SuDDDS scan [12, 10] maximizes a likelihood-ratio statistic for a declared-treatment DiD over candidate break quarters t_0 and window widths $W \in \{2, 4, 8\}$. Because the unit column is the chip itself, no subset search is available to the known-treatment scan: it localizes over (t_0, W) only, so *date localization is the contest* — does the maximum-LLR break land on the declared quarter? The primary model is Bernoulli with within-cluster calibration (wcc); a normal-model cross-check is calibrated under a dependence-preserving AR(1)-by-unit null. Permutation calibration uses $Q = 99$ draws at seed 0, so the smallest attainable scan p is 0.01.

Leg 2: unsupervised discovery rematch. The same scan machinery is then run with the treatment column deleted: θ is the observed flow, a categorical chip-name dimension is admitted to the subset search (model normal, method single_delta, restarts 8, panel-null $Q = 99$, seed 0), and the scan reports whatever subset $\times$ window is sharpest. Discovery never reads treatment labels; the question is whether the policy set is the panel’s sharpest structure or merely sits inside something larger.

Leg 3: effects. A manual two-way fixed-effects (TWFE) DiD with chip and quarter fixed effects gives $\hat{\tau}$ in asinh units with CR1 standard errors clustered by chip. With at most 7 clusters, CR1 t -statistics are known to over-reject [3] — and demonstrably do here — so the *inference of record* is an exact placebo-in-space test in the spirit of [5]: every same-size subset of control chips is pseudo-treated (true treated units dropped) and p is the rank of $|\hat{\tau}|$ in that exact distribution. The preferred specification is lifecycle-matched: donor chips alive (nonzero) in all 8 quarters, removing births and deaths from the donor pool. Robustness: dropping 2025Q1 (a visible pull-forward spike of 54.2k H100e before the April license), restricting to non-Chinese controls (substitution-contamination check), a placebo-in-time at pseudo- $t_0 = 2024Q4$ on pre-period data only, and an event study relative to 2025Q1.

Honest refusals, stated as run. natex’s built-in dd/synthetic-control/GESS effect estimators refused on all three panels (0 usable placebo units; ≥ 5 required) — hence the manual TWFE leg. The scan’s anticipation test refused on all known-treatment runs (no usable pre-quarters at these window widths; reported as not-passed with $p = \text{None}$, a refusal rather than a failure). And the lifecycle-matched Oct-23 effect leg refused outright: 0 of 5 control chips are alive in all 16 quarters, so no matched donor pool exists. Refusals are reported as refusals, not patched over.

4 Results

Localization: six for six, one significant. Every known-treatment scan puts the maximum-LLR break at the declared quarter exactly (Table 1, panel A). For the Oct-2023 round (7 chips $\times$ 16 quarters, treated A800+H800) the Bernoulli scan discovers $t_0 = 8095 = 2023Q4$ with $W = 4$ and $\text{LLR} = 4.896$, scan $p = 0.030$ ($Q = 99$); the normal/ar1_unit cross-check finds the same t_0 and W with $\text{LLR} = 4.085$, $p = 0.14$. The banned SKUs’ shipments go to literal zero within one quarter (Figure 1a). For the H20 round the localization is equally exact but never significant: sales panel $p = 0.44$ (Bernoulli) and 0.60 (normal); components panel (treated H20+MI308X) $p = 0.13$ and 0.23. The composition check passes ($p = 1.0$) in all six scans, and role-assignment sanity passes throughout: the scans are dating the declared treatment, not rediscovering an artifact.

Discovery: the generation, never the policy set. With treatment labels removed (Table 1, panel B), the Oct-23 panel’s sharpest structure is {A800, H100/H200, H800, TPU v4, TPU v5e} at 2022Q4 (LLR = 16.16, $p = 0.01$ floor) — the Hopper production ramp and the birth of the A800, itself the workaround SKU created by the October-2022 round. The ban quarter 2023Q4 does not appear in the top five discoveries; the closest is {A100, A800, H800} at 2024Q1 (rank 3) — the banned SKUs plus the A100’s end-of-life. On the H20 sales panel, discovery puts the sharpest break at 2025Q2 *exactly* (LLR = 17.32, $p = 0.05$), but the discovered subset is {H100/H200, H20, MI300A, TPU v5e, TPU v5p, Trainium1} — the H20 merged into the entire Hopper-generation wind-down, with the complementary Blackwell-generation upshift as the runner-up. The components panel repeats the pattern one quarter early (7-chip old-generation set at 2025Q1, LLR = 16.41, $p =$

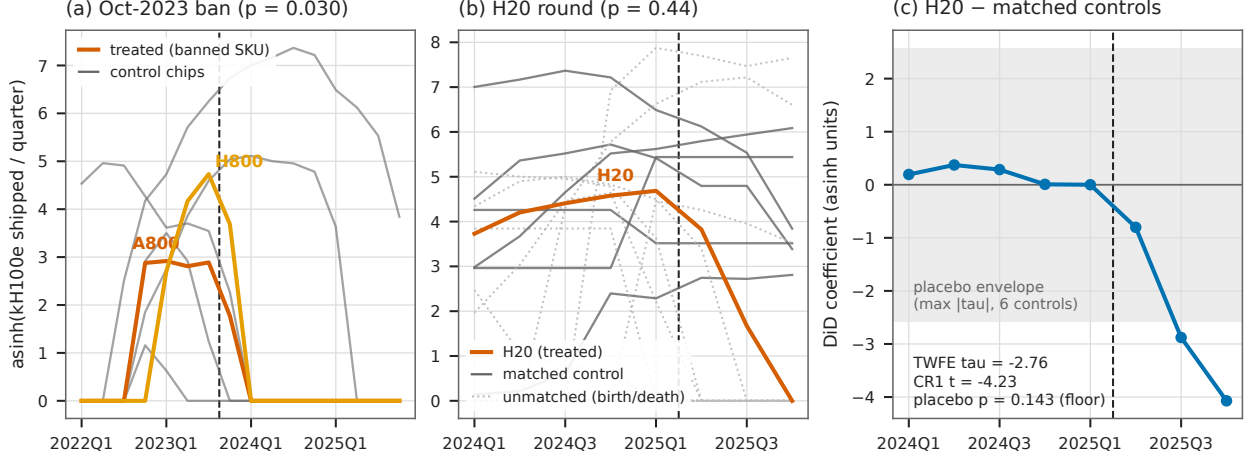


Figure 1: (a) Oct-2023 round: asinh shipment flows for the banned SKUs (A800, H800) against the five incumbent control chips; the scan puts the break exactly at the declared quarter (dashed line, 2023Q4 boundary) with $p = 0.030$, and treated shipments hit literal zero within one quarter. (b) H20 round: the H20 against 6 lifecycle-matched controls (solid) and 13 unmatched chips entering or exiting the market (dotted) — the churn that defeats attribution; the break again lands exactly on the declared quarter but at $p = 0.44$. (c) Event study, H20 minus matched controls relative to 2025Q1: flat pre-trend, monotone post-collapse to -4.07 asinh units by 2025Q4; the shaded band is the placebo-in-space envelope (max $|\tau|$ over the 6 matched controls), which the pooled estimate exceeds only at the $p = 1/7$ floor.

0.02). Every autonomously discovered subset is a chip *generation*, never the policy set: at product granularity, the H20 ban and the Hopper-to-Blackwell transition are the same event time.

Effects: large, sign-correct, unattributable. The lifecycle-matched TWFE estimate for the H20 (Table 1, panel C) is $\hat{\tau} = -2.756$ asinh units (CR1 se 0.652, $t = -4.23$) with 0 of 6 matched placebos exceeding — but 6 controls cap the exact test at its $p = 1/7 = 0.143$ floor. The event study is clean in shape: pre-period coefficients $+0.19, +0.37, +0.29, +0.01, 0$; post-period $-0.80, -2.88, -4.07$ (Figure 1c). The naive (unmatched) legs reproduce the lifecycle failure and calibrate how oversized CR1 is on these panels: the naive H20 spec has $t = -4.09$ yet 7 of 19 placebo chips produce larger pseudo-effects ($p = 0.400$); naive Oct-23 gives $p = 0.727$; naive components $p = 0.203$. Robustness on the matched leg: dropping the 2025Q1 anticipation spike leaves $\hat{\tau} = -2.799$ ($t = -3.73$) but costs one placebo exceedance ($p = 0.286$); restricting to the 3 non-Chinese controls gives $\hat{\tau} = -2.018$ with placebo $p = 0.750$ — the contamination check fails its placebo test, with the H100/H200’s own Blackwell wind-down (-2.12) and Trainium2 ($+2.88$) both comparable to the estimate. The placebo-in-time is clean ($\hat{\tau} = -0.280$, $t = -0.67$).

5 Caveats and Conclusion

Four caveats bound the claims. First, *product-lifecycle churn is a competing sharp treatment*: unsupervised discovery finds only chip-generation subsets at the policy quarters — the Hopper wind-down with the H20 merged into it at 2025Q2, the Hopper ramp with the A800’s birth at 2022Q4 — so at product granularity the H20 ban and the Blackwell transition are colinear in event time and cannot be separated inside this dataset. Second, *control contamination*: 3 of the 6

Table 1: Headline numbers of record, all three legs. Panel A: known-treatment SuDDDS scans; the declared quarter is 2023Q4 for the Oct-23 panel and 2025Q2 for both H20 panels, and $\hat{t}_0$ is the scan’s maximum-LLR break quarter. Panel B: unsupervised discovery (no treatment column; $\theta =$ the flow). Panel C: TWFE effects in asinh units; the exact placebo-in-space p is the inference of record.

<i>Panel A: known-treatment scans (date localization)</i>				
	$\hat{t}_0$	W	LLR	scan p
sales Oct-23 (7×16 ; A800+H800), Bernoulli/wcc	2023Q4 exact	4	4.896	0.030
normal/ar1_unit cross-check	2023Q4 exact	4	4.085	0.14
sales H20 (20×8 ; H20), Bernoulli/wcc	2025Q2 exact	4	1.532	0.44
normal/ar1_unit cross-check	2025Q2 exact	2	0.914	0.60
components H20 (15×8 ; H20+MI308X), Bernoulli/wcc	2025Q2 exact	4	2.331	0.13
normal/ar1_unit cross-check	2025Q2 exact	2	1.949	0.23
<i>Panel B: unsupervised discovery (top subset; every one a chip generation)</i>				
	$\hat{t}_0$	W	LLR	p
sales Oct-23: {A800, H100/H200, H800, TPU v4, TPU v5e}	2022Q4	4	16.16	0.01 (floor)
sales H20: {H100/H200, H20, MI300A, TPU v5e/v5p, Trainium1}	2025Q2 exact	4	17.32	0.05
components H20: 7-chip old-generation set (incl. H20, MI308X)	2025Q1	4	16.41	0.02
<i>Panel C: TWFE effects (asinh units; exceedances = placebos $\geq \hat{\tau}$)</i>				
	$\hat{\tau}$	CR1 se	t	exceed., p
matched sales H20 (6 controls alive all 8 q)	-2.756	0.652	-4.23	0/6, 0.143 (floor)
drop 2025Q1 (anticipation spike)	-2.799	0.751	-3.73	1/6, 0.286
non-Chinese controls only (3)	-2.018	0.939	-2.15	2/3, 0.750
naive sales Oct-23 (all 5 controls)	-1.070	1.160	-0.92	7/10, 0.727
naive sales H20 (all 19 controls)	-2.347	0.574	-4.09	7/19, 0.400
naive components H20 (all 13 controls)	-2.059	0.876	-2.35	15/78, 0.203
placebo-in-time (pseudo- t_0 2024Q4, pre-only)	-0.280	0.420	-0.67	clean
matched Oct-23 analog	refused: 0 of 5 control chips alive all 16 quarters			

Notes: all runs **natex** v0.2.0, seed 0. Scan p -values from permutation calibration with $Q = 99$ (floor 0.01); Bernoulli scans use within-cluster calibration (wcc), normal scans a dependence-preserving AR(1)-by-unit null. Composition checks pass ($p = 1.0$) in all six panel-A scans; the anticipation test refused (no usable pre-quarters). CR1 clusters by chip; with $G \leq 7$ the exact placebo-in-space p is the inference of record, and the naive H20 row ($t = -4.09$, placebo $p = 0.400$) calibrates CR1’s oversizing on these panels. Matched controls: Ascend 910B, Ascend 910C, H100/H200, MI300X, Siyuan 590, Trainium2 (placebo $\hat{\tau}$: -1.03, +2.06, -2.58, -1.50, +1.63, +1.42).

lifecycle-matched controls are Chinese accelerators (Ascend 910B/910C, Siyuan 590) that plausibly *gained* demand from the ban — a SUTVA violation biasing $\hat{\tau}$ negative (their placebo $\hat{\tau}$ ’s are +2.06, +1.63, -1.03) — while the no-China robustness leg (3 controls) gives $\hat{\tau} = -2.02$ with placebo $p = 0.75$: the matched leg’s 0/6 leans on contaminated donors. Third, *measurement*: Huawei and Cambricon flows are annual allocations smeared over quarters — smooth by construction, understating placebo dispersion — and CR1 t -statistics are demonstrably oversized here (the naive H20 leg pairs $t = -4.09$ with placebo $p = 0.400$), which is why the exact placebo-in-space p is the inference of record. Fourth, *resolution floors*: 6 matched controls cap the placebo test at $p = 1/7 = 0.143$ and the scan permutation floor is 0.01 ($Q = 99$); no p -value below its floor is claimed. Relatedly, all treated units share one policy date per round, so scan p -values come from the fitted-null panel calibration rather than cross-unit timing variation — the Bernoulli $p = 0.030$ for Oct-23 is the defensible headline, with the normal cross-check ($p = 0.14$) reported beside it.

The verdict is deliberately asymmetric. Date localization at product granularity *works*: all six known-treatment scans land on the declared quarter exactly, the October-2023 round clears its calibrated null ($p = 0.030$), and the banned SKUs’ shipments hit literal zero within one quarter — a policy written against named products is visible in product-level data on the dot. Attribution of

effect *size* at this granularity does not work, and the pipeline says so rather than manufacturing a significant number: the H20 collapse is large ($\hat{\tau} \approx -2.8$ asinh units) and correctly signed, but placebo chips generate ban-sized effects from ordinary lifecycle churn, the clean-donor robustness fails its own placebo test, and the matched Oct-23 leg refuses for want of donors. For an automated natural-experiment pipeline, the refusals and the correctly identified confound are the point: the machinery distinguished what these panels can establish — *when* — from what they cannot — *how much*.

Reproducibility. All estimates were produced with `natex` v0.2.0 [12] at seed 0 from frozen extracts of the Epoch AI datasets [8, 7] (not committed to the repository). Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline scan and effect estimates against the numbers of record before drawing.

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